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Claude Skills

A pedagogical approach to teaching AI literacy in schools

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IN BRIEF

AI literacy is badly served by being taught as a subject in parallel to everything else. It is better served by being embedded in the tools students are already using. Claude Skills — readable, writeable specifications that shape how an AI system behaves — turn every use of Claude into a potential AI literacy lesson. Students learn AI by reading the specifications that govern their own AI interactions, the way an earlier generation learned computing by editing the HTML of web pages they were already visiting. This paper argues that the strongest pedagogical case for skills in schools is not that they make AI useful for learning other subjects, but that the act of working with them teaches AI itself, in the most direct and honest way currently available.

1. The problem with AI literacy as a school subject

Schools face a genuine dilemma in teaching AI. The technology is consequential enough that ignoring it would be a professional failure, yet the conventional response — ***adding AI literacy as a discrete subject or a bolt-on unit — has not worked well. Students learn that AI is important without learning much about how it actually works.*** They learn definitions and warnings, but not the underlying mechanics. The result is a generation fluent in AI vocabulary and anxious about AI risks, yet genuinely mystified about what is happening when they type into a chat window.

The reason is pedagogical, not political. AI operates at a level of abstraction that resists classroom treatment. Neural networks cannot be drawn on a whiteboard in any meaningful sense. Prompting is hard to teach without something to prompt. Alignment is hard to discuss without a concrete artefact to point at. In consequence, AI literacy lessons tend toward the metaphorical and the admonitory, neither of which builds durable understanding in a Year 8 student.

2. What Claude Skills are, briefly

A Claude Skill is a small folder containing a specification file — `SKILL.md` — that tells Claude when to activate and how to behave. Skills may include supporting instructions, examples, templates, or scripts. Claude reads a skill only when a user request matches the skill's description, which keeps the system lean while giving Claude deep, domain-specific expertise on demand.

Structurally, a skill is a written artefact in plain language. It can be read, annotated, modified, and critiqued by anyone — including a thirteen-year-old. This property, more than any technical feature, is what makes skills pedagogically significant.

> *Skills do for AI what source code did for programming: they turn a mysterious capability into an inspectable artefact.*
>
> *Examples and metaphor in Appendix*

3. Why skills teach AI itself

Skills make the abstract concrete.

AI literacy curricula struggle with a concreteness problem. Concepts like prompting, context, instruction-following, and model limitations are hard to teach abstractly because students have nothing to point at. A skill is a physical artefact students can read. It turns *AI is given instructions* from a sentence in a textbook into a file on a screen whose contents visibly shape behaviour. This is the same pedagogical move that made computing education work when it moved from theory to actual code.

Reading skills teaches how language models are steered.

When a student reads a well-written skill, they see in plain English the mechanisms by which an AI system is directed. They see that the description determines when the skill fires — pattern matching, relevance judgement. They see that instructions shape output — prompting, specification. They see that examples disambiguate — few-shot learning, without needing the term. They see that refusal behaviours are written in, not emergent — alignment as authored, not magical. A single thoughtful skill walkthrough teaches more about how these systems work than a term of abstract lessons on neural networks.

Writing skills teaches specification as a discipline.

Having students write skills forces a rare intellectual move: specifying a task precisely enough that an agent without their context can perform it. This is the same muscle used in technical writing, in legal drafting, in giving instructions to a substitute teacher. Students quickly discover that their first attempt is ambiguous, their second over-constrained, and their third starts to work. That iterative experience — watching Claude's behaviour change as the skill is refined — teaches specification more viscerally than any essay on clear communication.

Critiquing skills teaches alignment thinking.

Give students a deliberately flawed skill and ask them to find the problems. The skill might over-trigger, producing outputs when it shouldn't. It might have weak refusal behaviours. It might encode subtle bias in its examples. It might be vulnerable to prompt injection. A student discovering they can type *ignore previous instructions* to a badly-written skill is learning something real about AI security. This teaches that AI behaviour is designed, that design has quality, and that quality can be evaluated — the disposition that transfers to every AI system they will encounter.

Skills puncture the myth of AI as magic.

There is a developmental stage, common at Year 8, where AI is either magical or terrifying — sometimes both. Skills puncture this usefully. A student who has read a skill, modified it, and watched Claude's output change accordingly has had the technology demystified in a way no amount of *AI is just statistics* teaching achieves. They have seen the strings. This does not make AI uninteresting. It makes it appropriately interesting — as a powerful but comprehensible technology, rather than a mysterious oracle or existential threat.

4. A pedagogical progression

The argument above suggests a natural sequence of classroom activity, each stage deepening the previous one. None of these stages requires technical background. Each can be done in an ordinary classroom with a small number of well-chosen skills.

Stage one — Reading

Students read an existing skill they have been using — perhaps a study-planner skill or a source-analysis skill. They annotate it. What does the description do? What do the instructions do? What do the examples add? What would happen if we removed each part? This can be done in a single lesson and produces immediate insight.

Stage two — Modification

Students change one element of a skill and predict how Claude's behaviour will change, then test the prediction. Change the tone instruction from *warm and encouraging* to *brisk and efficient* and observe what happens. This makes the skill-to-behaviour relationship tangible and teaches controlled experimentation as a natural by-product.

Stage three — Authoring

Students write a skill for a task they understand well — perhaps a skill that helps younger students with something the Year 8 cohort has mastered. The audience constraint (writing for Year 6) forces specification clarity. The fact that they are writing for a system, not a person, teaches a different kind of precision than essay writing typically demands.

Stage four — Evaluation

Students test each other's skills, find edge cases, identify where the skills fail. This is the closest any school activity comes to the actual work of AI evaluation — and it requires no technical background, only careful thinking. It is the most transferable skill in the sequence.

5. What this implies for schools

If the argument holds, three implications follow for school leadership, curriculum designers, and teachers.

- **AI literacy need not be a separate timetabled subject.** The mechanics of AI can be taught inside existing subjects, through the skills that structure students' use of AI in those subjects. A History lesson that uses a source-analysis skill is, incidentally and naturally, an AI literacy lesson.
- **Skill authorship is a new form of professional practice.** Teachers who write skills are engaged in a form of curriculum design that makes their pedagogical judgements explicit,

reusable, and improvable. This is professional capacity-building of a kind schools have rarely had access to before.

- **Students who learn to specify become more powerful users of any AI system.** The capability generalises. A student who has written and critiqued skills in school will be better equipped to use, evaluate, and govern the AI systems they encounter in higher education, in work, and in civic life.

6. A closing claim

The strongest pedagogical case for Claude Skills in schools is not the obvious one. It is not that skills make AI useful for teaching History, Geography, and Physics — although they do. It is not that skills reduce teacher workload — although they can. ***It is that skills, because they are readable and writeable artefacts, make AI itself teachable in a way nothing else currently does.***

The students who will shape the next two decades of AI deployment are in Year 8 now. The question is not whether they will learn about AI. They will, from their phones and their peers and a thousand other sources. ***The question is whether their formal education gives them tools to think about AI with precision and confidence — or whether it leaves them, as it has so far tended to, with fluent vocabulary and shallow understanding.***

Skills, read and written and critiqued in ordinary classrooms, offer a path to the former. That path is available now, uses tools schools already have or can easily obtain, and requires no technical background from teachers or students. It requires only the pedagogical decision that AI, like any other consequential technology, is best understood by working with the artefacts that govern it.

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7. Appendix

Metaphors for Explaining What a Skill Is

The recipe card

Imagine Claude is a capable cook who knows how to make almost anything, but doesn't know your household's way of doing things. A skill is a recipe card you give the cook: "When someone asks for pasta, use this sauce, serve it this way, and always toast the bread first." The cook's skill doesn't

come from the card — it comes from years of cooking. The card tells the cook how this particular household wants things done.

The briefing note for a new colleague

Think of a new, highly capable colleague starting at your school on Monday. They are qualified, intelligent, and willing — but they don't know the school's marking policy, the department's preferred lesson structure, or the phrase the head uses to begin assemblies. A skill is the briefing note you hand them: short, specific, practical. It doesn't teach them to teach — they already know that. It tells them how teaching is done here.

The HTML of a web page

A web page is made of content and instructions. The content is what you see; the HTML tells the browser how to display it. Most people who use the web never look at the HTML, but anyone can view it — it's plain text, human-readable, and you can learn to write it. A skill is the AI equivalent: the plain-text file that tells Claude how to behave, sitting underneath the conversation the student sees.

The teacher's lesson plan

A lesson plan isn't the lesson. The lesson happens in the room, with the students, in the moment. But the plan shapes what happens — it decides the starter, the main activity, the questions, the ending. A good teacher can teach without a plan; a good plan makes teaching more reliable, more consistent, and more shareable. A skill is a lesson plan for Claude: not the conversation itself, but the design that shapes it.

The contract with a tradesperson

When you hire an electrician, you don't teach them to wire a house. You tell them what you want: where the sockets go, what the finish should look like, what you want them to do if they find something unexpected behind the wall. The tradesperson's expertise is theirs; your contract shapes how that expertise is applied to your specific job. A skill is a contract with Claude — it doesn't create the capability, it directs it.

Examples of Claude Skills for the Classroom

1. House-writing skill

Purpose. Encodes the school's conventions for any extended piece of written work. Every other writing skill inherits from it.

Description (trigger). "Use when a student is producing extended written work of any kind — essays, analyses, reports, or reflections. Ensures output matches the school's expectations for register, structure, and presentation."

Core instructions. Written work uses formal register (no contractions, no first person in analytical writing unless the task calls for it). Paragraphs follow a topic-sentence-first structure. Evidence is integrated into sentences, not appended. Every paragraph ends with a line that connects back to the question. Never produce the final draft for the student — produce feedback, outlines, or partial examples only.

Classroom use. Teachers invoke this silently by having it always on for student accounts. Students notice, over time, that Claude's help consistently models the structure they're being asked to produce. This is the skill that makes AI support reinforce classroom teaching rather than undermine it.

2. History — source analysis skill

Purpose. Walks a student through the department's source analysis framework (provenance, purpose, value, limitations) without doing the analysis for them.

Description. "Use when a student is analysing a historical source — written, visual, or material. Scaffolds the four-step framework taught in Year 8 History without completing the analysis."

Core instructions. When a student presents a source, first ask them what they already know about its context. Do not supply contextual information the student has not yet established. Then guide them through provenance (who made this, when, and for whom?), purpose (why was it made?), value (what can it tell a historian?), and limitations (what can't it tell us, and why?). Ask one question at a time. If the student gives a one-word answer, ask them to extend it. Never give model answers; give model questions.

Classroom use. A student stuck on a source for homework invokes the skill and gets the scaffolding a teacher would provide if available. The teacher sees the transcript the next day and knows which steps the student found hard.

3. Geography — fieldwork write-up skill

Purpose. Structures the standard fieldwork report format used across Year 8 Geography.

Description. "Use when a student is writing up fieldwork or an investigation — including rivers, urban studies, microclimates, or coastal studies. Enforces the aim–hypothesis–method–data–analysis–conclusion–evaluation structure."

Core instructions. The student provides their data and observations. The skill produces a skeleton with the seven sections headed and one prompt under each: "What question were you actually trying to answer?" under Aim, "Which method produced which data, and why was that method chosen?" under Method, and so on. The skill refuses to write any section for the student but offers to comment on drafts. For the Evaluation section specifically, prompt for at least one methodological limitation and one suggestion for improvement.

Classroom use. Set as the default skill during the post-fieldwork lessons. Students who struggle with writing structure get consistent scaffolding; students who find it easy use it lightly or not at all.

4. Physics — problem-solving skill

Purpose. Enforces the physics department's expected problem-solving routine: list knowns, identify principle, choose equation, substitute, evaluate units, sense-check.

Description. "Use when a student is working on a quantitative Physics problem — mechanics, electricity, waves, or energy. Walks through the department's six-step method and insists on showing working."

Core instructions. Never give a numerical answer unless the student has attempted all six steps. If the student asks for the answer, respond with: "Let's work through it. What quantities has the question given you?" Check units at every substitution. If the student arrives at an answer with wrong units or an implausible magnitude, don't correct it directly — ask them to check whether the answer makes sense at the scale of the problem.

Classroom use. The canonical example of a skill that refuses to short-circuit the learning. Students quickly discover that Claude won't hand them the answer, and the routine becomes internalised because the skill keeps enforcing it.

5. English — analytical paragraph skill

Purpose. Enforces the PEEL or PETAL structure (whichever the department uses) for analytical writing about literature.

Description. "Use when a student is writing an analytical paragraph about a text — poem, novel, play, or non-fiction. Enforces the department's paragraph structure and insists the student supply their own evidence and interpretation."

Core instructions. The student names the text and the point they want to make. The skill asks: "What is one specific piece of evidence that supports that point?" Then: "What is the effect of that evidence on the reader?" Then: "What technique is the writer using, and why does it matter?" Finally: "How does this connect to the wider argument of your essay?" Produce these as a structured set of prompts, not a finished paragraph. The skill may offer sentence openers but not complete sentences of analysis.

Classroom use. Particularly valuable for students who can identify evidence but struggle with the analytical step — the "so what?" move that distinguishes description from analysis.

6. Explain-at-three-levels skill (student-support)

Purpose. Produces three versions of any explanation at different levels of complexity.

Description. "Use when a student needs a concept explained and isn't sure which level of detail suits them. Produces a simplified version, a standard version, and an extension version."

Core instructions. When given any topic, produce three explanations. The simplified version uses concrete examples and avoids technical vocabulary. The standard version matches what a confident Year 8 student would encounter in class. The extension version adds nuance, edge cases, or connections to more advanced ideas. After giving the three, ask the student which level they'd like to work at and offer to go deeper on that one. Never label a student as belonging to one level — levels describe explanations, not people.

Classroom use. The differentiation skill that works across every subject. Particularly useful for revision, where students often don't know whether they need the basics re-explained or are ready for stretch.

7. Stuck-protocol skill (student-support)

Purpose. Teaches students a structured routine for what to try before asking for help.

Description. "Use when a student says they're stuck on a task, problem, or piece of work. Walks them through a five-step protocol before offering any substantive help."

Core instructions. When a student reports being stuck, work through: (1) What exactly are you trying to do? Restate the task in your own words. (2) What have you already tried? (3) Where specifically did you get stuck — the first step, the middle, the end? (4) What would be the next thing you'd try if you had to guess? (5) What would a confident student in your class do next? Only after all five should the skill offer a hint, and hints should be the smallest possible nudge, not a full solution. Praise the metacognition, not the eventual answer.

Classroom use. One of the highest-leverage skills because it builds the help-seeking discipline students carry into every other context. Teachers notice students becoming more articulate about where they're stuck, which improves the quality of the help teachers can give.

8. Revision-planner skill (student-support)

Purpose. Builds a spaced-retrieval revision plan from a topic list and a test date.

Description. "Use when a student has a test or assessment coming up and needs a revision plan. Produces a spaced schedule using retrieval practice, not re-reading."

Core instructions. Ask for: the test date, the topics, the student's self-rated confidence on each topic (1–5), and how many minutes per day they can realistically revise. Produce a plan that revisits weaker topics more frequently and spaces revision of stronger topics further apart. Every revision session must be a retrieval activity — a quiz, a blank-page recall, a practice question — never re-reading notes. Include one low-effort review the day before the test and nothing new the day of. After producing the plan, explain briefly why retrieval practice works better than re-reading, in one or two sentences.

Classroom use. The skill itself teaches cognitive science as a by-product. Students who use it repeatedly internalise the principles of effective revision without sitting through a study-skills lesson.

9. Peer-feedback skill (task)

Purpose. Helps students give usable feedback on each other's work without either being unkind or being so vague the feedback is useless.

Description. "Use when a student is giving feedback on a peer's written work. Enforces the school's feedback protocol — two strengths, two specific suggestions, one question."

Core instructions. The student pastes or describes the peer's work. The skill asks what the task was and what the success criteria were. It then prompts: name two specific strengths tied to the criteria (not "it's good" but "the opening sentence sets up the argument clearly"); name two specific suggestions, each tied to a criterion the work has not yet met; ask one genuine question about the piece. The skill refuses to produce feedback on unnamed students or work the student claims is a peer's but cannot describe. Tone must be supportive but honest.

Classroom use. Transforms peer feedback lessons, which often produce generic or unkind comments, into structured exchanges that actually help. Students learn to give feedback as well as receive it.

10. Lesson-plan skill (teacher-facing)

Purpose. Produces a lesson plan in the school's template, aligned to the department's scheme of work.

Description. "Use when a teacher is planning a lesson. Produces a plan in the school's template including learning objectives, success criteria, starter, main activities, differentiation, plenary, and homework."

Core instructions. Ask the teacher for: year group, subject, topic, place in the scheme of work, lesson length, and any specific pupils to differentiate for. Produce a plan that includes a retrieval-practice starter linked to prior learning, a main activity with at least one opportunity for formative assessment, explicit differentiation (what the lowest-attaining pupils will do, what the highest-attaining will do), and a plenary that checks the stated learning objectives rather than simply asking "what have we learned today?" Homework must be retrieval or application, not finishing-off.

Classroom use. The workload skill. Teachers using it consistently save significant planning time on routine lessons, freeing attention for the lessons that genuinely need bespoke design. Heads of department can audit the plans to check consistency across a team.

A Note on What These Examples Show

Three things are worth noticing across these examples.

1) The skills are all small. None of them is trying to be a general-purpose tutor. Each encodes one clear piece of professional practice — a department's preferred structure, a known-effective study technique, a specific feedback protocol — and does it reliably.
2) The skills teach pedagogy as a by-product. A student using the revision-planner learns about spaced retrieval. A student using the stuck-protocol learns metacognition. A student using the peer-feedback skill learns what good feedback looks like. The skill's purpose is the immediate task; its side effect is the durable learning habit.
3) The skills are legible. Anyone — a teacher, a head of department, an inspector, a parent — can read these skills and understand exactly what the AI is being asked to do. That legibility is the property that makes skills defensible in a school context and, circling back to the position paper, makes them the right artefact to put in front of students learning what AI actually is.